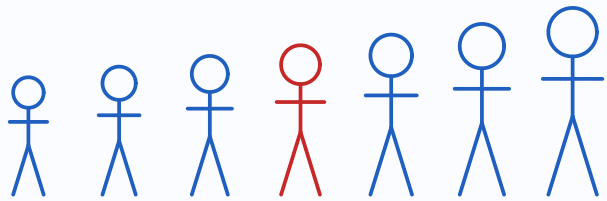
Chapter 24

Statistics: averages and spread

three numbers that tell the whole story



line them up by height:
the one in the middle is the median

TYPICAL?

HOW FAR?

HOW SPREAD?

Lecture 4 starts from a fact about brains: **we can only work with about four things at once**. A data set of hundreds of values is useless until we squash it into three or four summary numbers:

the average (what number is typical?), the range (what are the outer limits?) and the variance (how spread out are the values?).

Goes with **Lecture 4** (mean, median, mode, quartiles, box plot, variance, standard deviation, sample vs population).

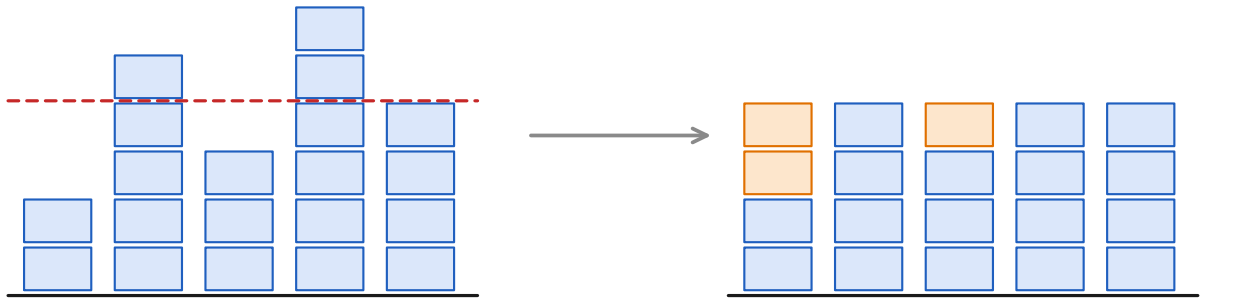
The mean: level the towers

RULE: the mean

mean $= \dfrac{\text{sum of the values}}{\text{number of values}}$, written $\bar{x} = \dfrac{\sum x}{n}$.

Turned round: **total = number of values × mean**. Every "find the missing value" question starts here.

Five friends have 2, 5, 3, 6 and 4 sweets. The **mean** is what each would get if they **shared them out equally**. Picture towers of blocks and move blocks from tall towers to short ones until they're level:



before: 2, 5, 3, 6, 4 — *after: every tower is 4 (orange = moved)*

Sharing out equally is the same as: **add them all up, divide by how many there are.**

$$\bar{x} = \frac{2+5+3+6+4}{5} = \frac{20}{5} = 4$$

Lecture 4 writes it with the sigma (adding-up) sign of Chapter 17:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$\bar{x}$ ("x bar") = the mean

$\sum x_i$ = add up every value $x_1, x_2, \ldots x_n$

n = how many values there are

The lecture's giant example: UK GDP is about 2.5 trillion dollars and there are about 65 million people, so the mean income is $2.5 \times 10^{12} \div (6.5 \times 10^{7}) \approx 38\,500$ dollars, about £24,000.

The median: the middle of the queue

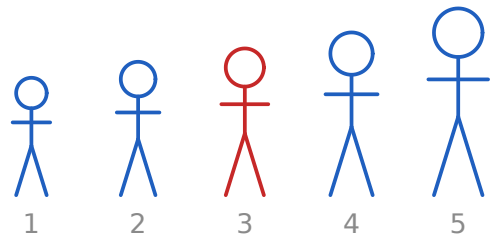
RULE: median and mode

Median: put the values **in order**; the median is the value in position $\frac{n+1}{2}$.

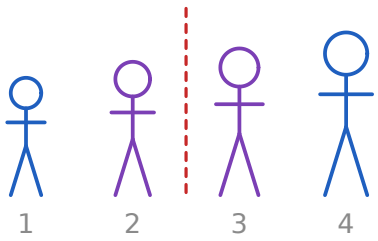
If n is even the position ends in .5: the median is halfway between the two middle values (add them and halve).

Mode: the value that occurs most often. There can be more than one mode, or none.

Line everyone up in order of size. The **median** is the person standing in the middle.



5 people: position 3 is the middle



4 people: nobody is in the middle, so go halfway between 2 and 3

The position of the middle is $\frac{n+1}{2}$ (Lecture 4 writes $M = x_{(n+1)/2}$). For $[1, 2, 3, 4, 5]$ that's position 3, so $M = 3$. For $[1, 2, 3, 4]$ it's position 2.5, so halfway between the 2nd and 3rd values: $M = \frac{2+3}{2} = 2.5$.

The mode is the **most common** value: the tallest stack in a dotplot. In 3, 5, 5, 6, 8, 5, 2 the mode is 5. No formula, just counting. For measurements like wages, where nobody earns exactly the same, you first group them into brackets (Chapter 23), then the mode is the busiest bracket.

Missing values: work with the total

PROCEDURE: missing-value problems

1. Turn every mean you are given into a total: total = number of values × mean.
2. Subtract the values you know from the total: what's left is the missing value (or their sum).
3. If a value is added or removed, change the total **and** the count, then divide again: new mean = new total ÷ new count.
4. Check: add the values up and divide.

Example: the mean of five numbers is 8. Four of them are 6, 7, 9 and 10. Find the fifth.

$$\text{total} = 5 \times 8 = 40$$ ← *total = n × mean*

$$40 - (6 + 7 + 9 + 10) = 40 - 32 = 8$$ ← *take away the four you know*

Example: four values have mean 15. A fifth value, 25, is added. What is the new mean?

$$4 \times 15 = 60, \qquad \frac{60 + 25}{5} = \frac{85}{5} = 17$$ ← *new total ÷ new count*

Example: six values have mean 10. The value 20 is removed. What is the new mean?

$$6 \times 10 = 60, \qquad \frac{60 - 20}{5} = \frac{40}{5} = 8$$ ← *one fewer value*

Example: the mean of x, $2x$ and $3x$ is 12. Find x.

$$\frac{x + 2x + 3x}{3} = 12 \;\Rightarrow\; 6x = 36 \;\Rightarrow\; x = 6$$

total = 3 × 12 = 36

Example: four tests average 66. What must the fifth score be to bring the average up to 70?

$$5 \times 70 - 4 \times 66 = 350 - 264 = 86$$ ← *total needed − total so far*

✗ Common mistake: averaging two means

✗ $\dfrac{20+40}{2} = 30$

group A: 10 people, mean 20; group B: 30 people, mean 40. Combine the totals, not the means

✓ $\dfrac{10 \times 20 + 30 \times 40}{40} = 35$

Averages from a frequency table

RULE: frequency tables

f = how many times the value x occurs. The number of values is $n = \sum f$ (not the number of rows).

Mean $= \dfrac{\sum fx}{\sum f}$: multiply each value by its frequency, add these up, divide by the total frequency.

Median: position $\frac{n+1}{2}$ with $n = \sum f$. Keep a running total of f to see which value sits in that position. Mode: the value with the biggest f.

Grouped data (classes like 10-20): use each class's **midpoint** as x. The mean is then only an **estimate**, because the exact values are lost.

Example: the number of brothers and sisters of 20 students.

siblings x	frequency f	fx	running total of f
0	5	0 × 5 = 0	5
1	8	1 × 8 = 8	13
2	4	2 × 4 = 8	17
3	3	3 × 3 = 9	20
total	**Σf = 20**	**Σfx = 25**	

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{25}{20} = 1.25$$

$$\text{median position} = \frac{20+1}{2} = 10.5$$ ← between the 10th and 11th values

The running total says values 1 to 5 are 0s and values 6 to 13 are 1s, so the 10th and 11th are both 1: **median = 1**. The mode is also 1 (frequency 8).

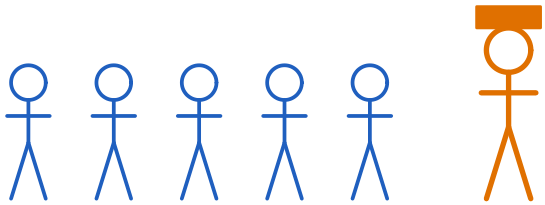
Example: grouped data: minutes spent on homework by 20 students.

minutes	frequency f	midpoint x	fx
0-10	4	5	5 × 4 = 20
10-20	10	15	15 × 10 = 150
20-40	6	30	30 × 6 = 180
total	**20**		**350**

$$\bar{x} \approx \frac{350}{20} = 17.5 \text{ minutes}$$ ← an estimate: midpoints stand in for the real values

A billionaire walks into a pub

Five people in a pub earn £20k, £22k, £25k, £28k and £30k. Mean £25k, median £25k. Then a billionaire earning £1,000 000k walks in:

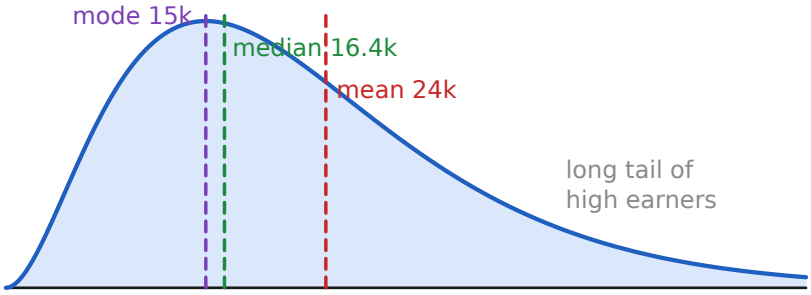


mean **£25k → £166 687k**

median **£25k → £26.5k**

the mean now describes nobody in the room

One extreme value (an outlier) drags the mean but barely moves the median. Data with a long tail on one side are called **skewed**, and UK wages are exactly like that. Lecture 4's three answers to "what is the average UK wage?":



The mega-rich few in the tail pull the mean right.

Median: usually the most honest for skewed data. Mean: best for symmetric (Normal) data. Mode: rarely used.

Lecture 4 also lists what's wrong with the wage data itself (part-time work not pro-rata, self-reporting, tax avoidance...), and hides a puzzle on each slide: a wrong title, a bad choice of graph (Chapter 23's case 1) and overlapping brackets like "£6000 to £7000" then "£7000 to £8000": which bracket does exactly £7000 go in?

Range, quartiles and the box plot

RULE: range, quartiles and IQR (Lecture 4's method)

Put the data in order. Median position $= \dfrac{n+1}{2}$.

Quartile position $= \dfrac{\text{median position} + 1}{2}$. Count that far **up from the smallest** for the lower quartile (LQ) and **down from the largest** for the upper quartile (UQ).

Range = largest − smallest. IQR = UQ − LQ. Semi-interquartile range = IQR ÷ 2.

RULE: fractional positions

Position 3.5 means **halfway** between the 3rd and 4th values: add them and halve.

In general, position $k + d$ (whole part k, decimal part d) means: start at the kth value and go the fraction d of the gap to the next one: $x_k + d\,(x_{k+1} - x_k)$.

So position 1.75 is **¾ of the way** from the 1st value to the 2nd, and 2.25 is ¼ of the way from the 2nd to the 3rd. Counting down from the top for the UQ, move the fraction **down** towards the next value.

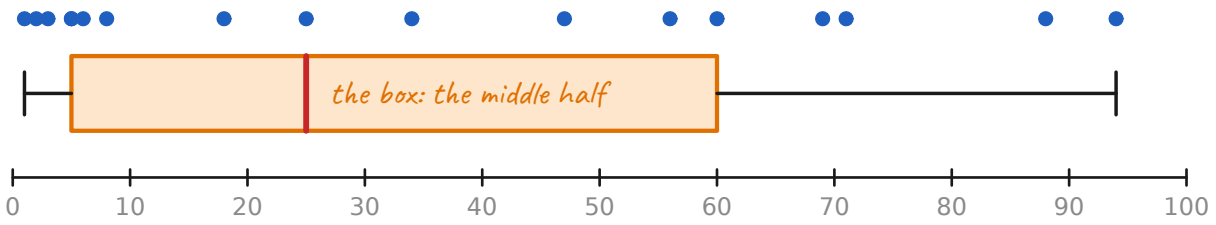
Calculators and software use slightly different quartile rules, so their answers can differ a little. Say which method you used.

Lecture 4's 17 values, in order. We'll find five landmarks and hang a box on them.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1	2	3	5	**5**	6	8	18	**25**	34	47	56	**60**	69	71	88	**94**
min				LQ				median				UQ				max

Median: position $\frac{17+1}{2} = 9$, so the 9th value: **25**.

Quartiles cut the data into quarters. Lecture 4: the quartile position is $\frac{\text{median position}+1}{2} = \frac{9+1}{2} = 5$. Count 5 from the bottom for the lower quartile (LQ = 5) and 5 from the top for the upper quartile (UQ = 60).



range	largest − smallest	**94 − 1 = 93**
interquartile range (IQR)	UQ − LQ: the width of the box	**60 − 5 = 55**
semi-interquartile range	half the IQR	**55 ÷ 2 = 27.5**

The range uses only the two most extreme values, so one outlier wrecks it. The IQR ignores the top and bottom quarters, so it's robust, like the median.

Example: quartiles with fractional positions: 2, 4, 6, 8, 10, 12, 14, 16 ($n = 8$).

$$\text{median position} = \frac{8+1}{2} = 4.5, \quad \text{median} = \frac{8+10}{2} = 9$$

halfway between the 4th and 5th

$$\text{quartile position} = \frac{4.5+1}{2} = 2.75$$

$$\text{LQ} = 4 + 0.75 \times (6 - 4) = 5.5$$ ← ¾ of the way from the 2nd value (4) to the 3rd (6)

$$\text{UQ} = 14 - 0.75 \times (14 - 12) = 12.5$$ ← 2.75th from the top: ¾ of the way from 14 down to 12

$$\text{IQR} = 12.5 - 5.5 = 7$$

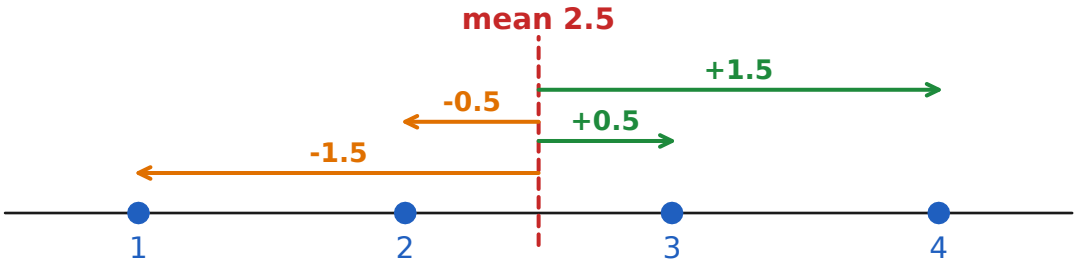
How spread out? A tug of war that always draws

RULE: deviation

The deviation of a value is its distance from the mean: $x - \bar{x}$ (negative below the mean).

The deviations of any data set **always add up to 0**, so their plain average is useless as a measure of spread.

Take $[1, 2, 3, 4]$: mean 2.5. But so does $[-1000, 608, 400, 2]$! The mean can't tell them apart; we need to measure how far values sit from the mean. That distance is the **deviation**: $x - \bar{x}$.



left pulls: −1.5, −0.5. Right pulls: +0.5, +1.5. Total: 0. Always!

Averaging the deviations is useless: the mean is the balance point, so the pulls to the left and right **always cancel to zero**. We need to get rid of the minus signs. Lecture 4's trick: **square them**.

Variance: the area of the average square

RULE: variance and standard deviation

Variance = the **mean of the squared deviations**. Standard deviation = $\sqrt{\text{variance}}$, back in the units of the data.

Whole **population** (N values, mean μ): $\sigma^2 = \dfrac{\sum(x-\mu)^2}{N}$, $\sigma = \sqrt{\sigma^2}$.

Sample (n values, mean $\bar{x}$): $s^2 = \dfrac{\sum(x-\bar{x})^2}{n-1}$, $s = \sqrt{s^2}$ (why $n-1$: three sections on).

Squaring a deviation means drawing a **real square** whose side is the deviation. Minus signs vanish (area is never negative), and far-away values get **much** bigger squares:



(-1.5)² = 2.25 (-0.5)² = 0.25 (+0.5)² = 0.25 (+1.5)² = 2.25

sum of squares = 2.25 + 0.25 + 0.25 + 2.25 = 5

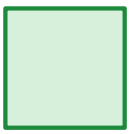
Problem: $[1, 1, 2, 2, 3, 3, 4, 4]$ is just as spread out, but has 8 squares, so its sum is 10. Bigger groups get bigger sums. **Fix: divide by the number of values.** That's the average square:

$$\frac{5}{4} = 1.25 \qquad \frac{10}{8} = 1.25$$ ← *same spread, same answer*

This average-square area is the **variance**, σ^2. Its side length (the square root) is the **standard deviation** σ, which is back in the same units as the data. Treating the values as a whole population, Lecture 4 writes the mean as μ and the count as N:

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

$$\sigma = \sqrt{1.25} \approx 1.118$$



area 1.25 = variance

side 1.118 = standard deviation

Units check (Lecture 4): $[10, 20, 30, 40]$ has variance 125, a hundred times bigger than 1.25, because squares of 10× longer sides have 100× the area. The standard deviation, $\sqrt{125} = 11.18$, is exactly 10× bigger: sensible. That's why we usually quote σ.

Wider range means more spread? Not always

Lecture 4's "range fallacy": $[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 5]$ has range 4, wider than $[1, 2, 3, 4]$ (range 3). Yet its variance is 1.06, **slightly less** than 1.25: thirteen values sit almost on top of the mean, and only one lonely square is big. **Range looks at two values; variance looks at all of them.**

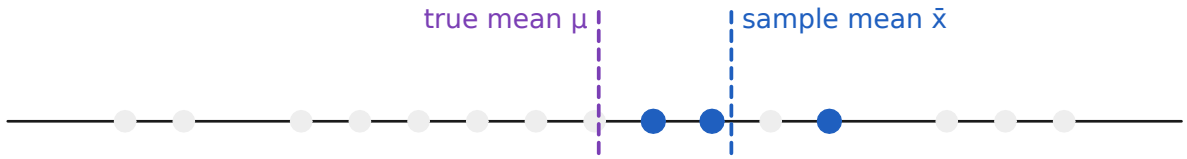
Sample or population, and the mysterious n − 1

A **population** is everyone you care about (all NTU students). A **sample** is the handful you actually measured (this class). Lecture 4 uses different letters so you always know which you mean:

	population (Greek)	sample (English)
mean	μ ("mew")	x̄ ("x bar")
standard deviation	σ ("sigma")	s
how many	N	n

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}} \qquad s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

Why $n - 1$ (Bessel's correction)? A sample huddles closer to **its own** mean than to the true population mean, because its mean was calculated from those very values. So its squares come out **too small**. Dividing by a slightly smaller number ($n - 1$) nudges the answer back up:



the 3 blue values are closer to x̄ than to μ, so their spread looks smaller than it really is

The fix matters most for small samples: dividing by 2 instead of 3 is a big change; by 999 instead of 1000 is tiny.

The recipe, one step per line

Lecture 4's "break it down" for the sample standard deviation of $[1, 2, 3, 4]$:

x	x − x̄	(x − x̄)²
1	-1.5	2.25
2	-0.5	0.25
3	+0.5	0.25
4	+1.5	2.25
		total 5

1 find the mean: 10 ÷ 4 = 2.5
2 subtract the mean from each x
3 square each one
4 add the squares: 5
5 divide by n − 1 = 3: 1.667
6 square root: s ≈ 1.291

Compare: dividing by $n = 4$ gave $\sigma = 1.118$ (treating the four values as the whole population). Calculators have both buttons: σ_n (or σx) and σ_{n-1} (or sx).

Shifting and stretching the data

RULE: adding or multiplying every value

Add a constant c to every value: the mean, median, mode and quartiles all go up by c. The spread does **not** change: range, IQR, variance and standard deviation stay the same.

Multiply every value by k: the mean, median, mode and quartiles are multiplied by k; the range, IQR and standard deviation by $|k|$; the variance by k^2.

Both at once ($y = kx + c$): new mean $= k\bar{x} + c$, new standard deviation $= |k| \times$ old.

Sliding every value along by 10 moves the whole data set, but the gaps between the values (and so the spread) are untouched. Tripling every value triples every gap.

Example: 2, 4, 6 has mean 4 and $\sigma \approx 1.633$.

$$\text{add } 10\text{: } 12, 14, 16 \;\Rightarrow\; \text{mean } 14,\ \sigma \approx 1.633$$

mean + 10, spread unchanged

$$\text{times } 3\text{: } 6, 12, 18 \;\Rightarrow\; \text{mean } 12,\ \sigma \approx 4.899$$ ← *both × 3*

Example: temperatures have mean 20 °C and standard deviation 5 °C. In Fahrenheit, $F = 1.8C + 32$.

$$\text{mean} = 1.8 \times 20 + 32 = 68\,^\circ\text{F}$$ ← *multiply and add*

$$\text{SD} = 1.8 \times 5 = 9\,^\circ\text{F}$$ ← *multiply only: the +32 doesn't spread anything*

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Mean	$\bar{x} = \dfrac{\sum x}{n}$ add them up, divide by how many
Total from a mean	$\text{total} = n \times \bar{x}$ the first step of every missing-value problem
Mean of a frequency table	$\bar{x} = \dfrac{\sum fx}{\sum f}$ grouped data: x = class midpoint (an estimate)
Median position	$\dfrac{n+1}{2}$ data in order; a .5 position means halfway between two values
Quartile position	$\dfrac{\text{median position} + 1}{2}$ count up from the bottom for LQ, down from the top for UQ
Fractional position	$k + d \;\rightarrow\; x_k + d\,(x_{k+1} - x_k)$ e.g. 1.75 = ¾ of the way from the 1st value to the 2nd
Mode	the most common value (or busiest class)
Range	$\text{largest} - \text{smallest}$
Interquartile range	$\text{IQR} = \text{UQ} - \text{LQ}$ semi-interquartile range = IQR ÷ 2
Population variance and SD	$\sigma^2 = \dfrac{\sum (x-\mu)^2}{N}, \qquad \sigma = \sqrt{\sigma^2}$ the whole population: mean μ, N values
Sample SD	$s = \sqrt{\dfrac{\sum (x-\bar{x})^2}{n-1}}$ a sample: mean $\bar{x}$, divide by $n-1$ (Bessel)
Add c to every value	averages + c; range, IQR, SD, variance unchanged
Multiply every value by k	averages × k; range, IQR, SD × |k|; variance × k²
Which average?	skewed data or outliers: median. Symmetric data: mean

Your turn

Data A: 3, 7, 7, 8, 10, 12, 15. Answers in grey.

Averages

1. Mean of A
2. Median of A
3. Mode of A
4. Add 100 to data A. Which average changes most?

1. 62 ÷ 7 ≈ 8.86 2. 8 3. 7 4. the mean (it becomes 20.25; median 9)

Spread

5. Range of A
6. Median position for 11 values? Quartile position?
7. Population variance and σ of 2, 4, 6
8. Sample s of 2, 4, 6

5. 12 6. 6th; (6 + 1)/2 = 3.5th from each end (halfway between the 3rd and 4th) 7. 8/3 ≈ 2.67, σ ≈ 1.63 8. √(8/2) = 2

Think

9. Why is median house price reported, not the mean?
10. Two classes both average 60%. What else would you want to know?

9. a few mansions skew the mean upward 10. the spread (standard deviation / IQR)

Find the unknown

11. The mean of 5 numbers is 8. Four of them are 6, 7, 9, 10. Find the fifth.
12. The mean of x, 2x and 3x is 12. Find x.
13. 4 values have mean 15. The value 25 is added. New mean?
14. 6 values have mean 10. The value 20 is removed. New mean?
15. Ten values have mean 7 and σ = 2. New mean and σ if every value is (a) increased by 5, (b) tripled?
16. x = 1, 2, 3 have frequencies 4, k, 2 and the mean is 1.8. Find k.

11. 8 (total 5 × 8 = 40; 40 − 32 = 8) 12. x = 6 (6x = 3 × 12 = 36) 13. 17 (60 + 25 = 85; 85 ÷ 5) 14. 8 (60 − 20 = 40; 40 ÷ 5) 15. (a) mean 12, σ = 2 (b) mean 21, σ = 6 16. k = 4 (Σfx = 10 + 2k = 1.8 × (6 + k))